



A novel antibody-TCR (AbTCR) T-cell therapy is safe and effective against CD19-positive relapsed/refractory B-cell lymphoma

Pengcheng He¹ · Haibo Liu¹ · Bryan Zimdahl² · Jie Wang¹ · Minna Luo¹ · Qi Chang² · Fangzhou Tian² · Fan Ni² · Duo Yu² · Huasheng Liu¹ · Limei Chen¹ · Huaiyu Wang¹ · Mei Zhang¹ · Stephan A. Grupp^{3,4} · Cheng Liu² 

Received: 19 April 2022 / Accepted: 10 June 2022 / Published online: 1 July 2022
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2022

Abstract

Purpose A barrier to widespread adoption of chimeric antigen receptor (CAR) T-cell therapy is toxicity. To address this, we recently developed a novel antibody-T-cell receptor (AbTCR) platform (trademarked as ARTEMIS[®]) which was designed to leverage natural immune receptor signaling and regulation. The AbTCR platform includes a gamma/delta ($\gamma\delta$) TCR-based AbTCR construct and a separate co-stimulatory molecule, both engineered to be tumor-specific. Here, we aim to assess the safety and preliminary efficacy of a CD19-directed AbTCR T-cell therapy.

Methods We generated ET019003 T cells, which are autologous CD19-directed AbTCR T cells. We then conducted an early phase I study to evaluate the safety and preliminary efficacy of ET019003 T cells for the treatment of CD19-positive relapsed/refractory (r/r) B-cell lymphoma.

Results Sixteen patients enrolled in this study and 12 patients were treated. Of the 12 patients treated, 6 patients (50%) achieved a complete response (CR), and 4 (33%) achieved a partial response (PR) (best objective response rate [ORR] of 83%). CRs were durable, including 2 patients with ongoing CRs for 22.7 months and 23.2 months. ET019003 was well-tolerated with an attractive safety profile. No patients experienced severe (grade ≥ 3) cytokine release syndrome (CRS) and only 1 patient experienced immune effector cell-associated neurotoxicity syndrome (ICANS) of any grade. Significant elevations of cytokine levels were not seen, even in patients with marked expansion of ET019003 T cells.

Conclusion This study provides initial clinical validation of the AbTCR platform as a novel cancer treatment with the potential to provide durable clinical benefit with low toxicity.

Trial registration NCT03642496; Date of registration: August 22, 2018.

Keywords Antibody-T-cell receptor therapy · T-cell therapy · B-cell lymphoma · Immunotherapy · CD19-directed therapy

Introduction

Engineered T cells are a promising approach for the treatment of many different types of cancer, particularly for the treatment of hematological malignancies that have become refractory to prior systemic therapy. The most widely used mode of engineered T-cell therapy is chimeric antigen receptor (CAR) technology (Barrett et al. 2014). United States Food and Drug Administration (FDA) approved CAR T-cell therapies targeting the CD19 antigen have induced complete remission of certain forms of cancer previously thought to have very poor prognosis (Neelapu et al. 2017; Maude et al. 2018; Schuster et al. 2019; Abramson et al. 2020; Wang et al. 2020). However, toxic side effects of CAR T-cell therapy, notably cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS)

Pengcheng He, Haibo Liu, and Bryan Zimdahl contributed equally to this study.

✉ Cheng Liu
cheng.liu@eurekainc.com

¹ Department of Hematology, The First Affiliated Hospital of Xi'an Jiaotong University, Xi'an, China

² Eureka Therapeutics, Inc., Emeryville, CA, USA

³ Department of Pediatrics, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, USA

⁴ Division of Oncology, Center for Childhood Cancer Research and Cancer Immunotherapy Program, Children's Hospital of Philadelphia, Philadelphia, PA, USA

are common, and in some cases can lead to life-threatening multiorgan failure (Turtle et al. 2016a; Gauthier and Turtle 2018; Park et al. 2018; Lee et al. 2019). The manifestation of these toxicities is likely due to the inherent structural design of the CARs. For example, the conventional second-generation CAR construct includes the direct fusion of the antigen recognition domain to the co-stimulatory (CD28 or 4-1BB) and CD3 ζ domains, which drives sustained T-cell activation and, often, subsequent release of large amounts of inflammatory cytokines (Brentjens et al. 2007; Maude et al. 2014, 2015).

Accordingly, there remains a need to develop new cellular therapies that can provide the potent anti-tumor cytotoxicity of CAR T cells but with reduced toxicities. As an alternative to the traditional CAR T-cell platform, we recently developed an antibody-T-cell receptor (AbTCR)-based cell therapy platform that links an antigen-binding antibody fragment (Fab) to the human gamma/delta ($\gamma\delta$) T-cell receptor (TCR) chains (Xu et al. 2018). We reasoned that because the AbTCR approach more closely resembles the endogenous TCR signaling machinery than do traditional CAR constructs, AbTCR T cells may generate potent T-cell-mediated tumor cytotoxicity with less cytokine-mediated toxicities. In pre-clinical data, T cells expressing an AbTCR construct targeting the CD19 antigen functionally match the potency of CAR T cells, but release significantly lower levels of cytokines upon the killing of target-positive cells both in vitro and in cell-line and patient-derived xenograft mouse models (Xu et al. 2018). To further augment AbTCR signaling, we subsequently optimized the AbTCR platform to include a co-expressed, yet separate, co-stimulatory molecule also designed to target a tumor-specific antigen. Importantly, unlike the linear CAR design of traditional CAR T-cell platforms, the configuration of this AbTCR system resembles the endogenous TCR/co-stimulatory receptor architecture in which co-stimulation is provided *in-trans* and acts as a potent synergistic signal that is naturally regulated (Chen and Flies 2013). The clinical product, ET019003 T cells, are autologous T cells genetically modified to co-express an AbTCR and a co-stimulatory molecule, both targeting the CD19 antigen (Online Supplemental Figure S1). Herein, we report the results of our early phase I clinical study of ET019003 T cells for the treatment of CD19-positive relapsed/refractory (r/r) B-cell lymphoma.

Methods

Trial design and patients

This was an exploratory, single arm, open-label, non-randomized early phase I clinical study to assess the safety and feasibility of administering autologous ET019003 T cells

to patients with CD19-positive relapsed/refractory (r/r) B-cell lymphoma. Preliminary exploration of the efficacy of ET019003 T cells in r/r B-cell lymphoma was also assessed. The study enrolled patients from November 2018 to January 2020 and was conducted at The First Affiliated Hospital of Xi'an Jiaotong University in China and was registered at www.clinicaltrials.gov as #NCT03642496. The Ethics Committee of The First Affiliated Hospital of Xi'an Jiaotong University reviewed preclinical data and approved the clinical protocol. All participants in the study provided written informed consent. The study was conducted in accordance with the principles of the Declaration of Helsinki. Patients were eligible if they had histologically confirmed CD19-positive r/r B-cell lymphoma. Previous therapy must have included at least one cycle of R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone) or a similar R-CHOP-like chemotherapy. Eligibility also required measurable disease as defined by at least one (1) measurable node of which the longest diameter (LDi) is greater than 1.5 cm or at least one (1) measurable extra nodal lesion of which the LDi is greater than 1.0 cm. An Eastern Cooperative Oncology Group (ECOG) performance status of less than or equal to two (2) was required. Additional trial information including the complete list of inclusion and exclusion criteria is provided in Online Supplemental Data.

Endpoints and assessments

The primary endpoints included safety and tolerability of ET019003 T-cell therapy and the cellular kinetics (i.e., expansion and persistence) of ET019003 T cells after infusion. For reporting of adverse events, the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE; version 5.0) was used, except for CRS, which was graded according to American Society for Transplantation and Cellular Therapy (ASTCT) consensus grading (Lee et al. 2019). Secondary endpoints included best overall response rate (ORR) and duration of response (DoR). The best ORR was defined as the combined percentage of patients who had a complete response (CR) or partial response (PR) at any time after initial infusion. DoR was defined as the time from the first documented disease response (CR or PR) to the date of first documented progression or death due to B-cell lymphoma. If a patient had not progressed or died at the time of data cutoff, DoR was censored on the date for the last available follow-up visit. Tumor assessments were conducted 1, 2, 3, 6, 9, 12, 18, and 24 months after initial infusion and response to treatment was assessed by the principal investigator (Pengcheng He) and radiologist according to the Lugano Classification 2014 (Cheson et al. 2014). By these criteria, a CR indicates the disappearance of measurable disease via computerized tomography (CT) scan or residual masses that are positron

emission tomography (PET)-negative. PR indicates at least 50% decrease in tumor burden with ongoing PET avidity.

ET019003 T-cell manufacture

For ET019003 T-cell manufacturing, peripheral blood mononuclear cells (PBMCs) were collected from patients by leukapheresis. PBMCs were isolated and purified from the leukapheresis product by Ficoll density centrifugation (GE Healthcare, Life Sciences, Shanghai). Anti-CD3/CD28 antibody coated beads (Thermo Fisher Scientific, Shanghai) were used for T-cell enrichment and activation. PBMCs were cultured in RPMI-1640 medium supplemented with 5% human serum, IL-7, and IL-15. Following T-cell activation, T cells were transduced with ET019003 Lentivirus (see Online Supplemental Data). Cells were expanded for an additional 8 days. Quality-control tests were performed on the final product before release (see Online Supplemental Data).

Lymphodepletion and ET019003 T-cell infusion

Before the first ET019003 T-cell infusion on day 0, patients received lymphodepleting (LD) chemotherapy consisting of cyclophosphamide at 250 mg/m²/day and fludarabine at 25 mg/m²/day for three days (median start date: day -5, range: day -7 to day -3). For safety, the first individual (patient BH05-P06) received an initial weight-adjusted dose of 1.5×10^6 ET019003 Receptor-positive T cells per kilogram of body weight (/kg). All other patients received an initial weight-adjusted dose of 3×10^6 ET019003 Receptor-positive T cells/kg. After the first dose, one or more weight-adjusted doses of 1.5×10^6 to 3×10^6 ET019003 Receptor-positive T cells/kg were permitted at the investigator's discretion based on an evaluation of the risks and benefits. Prior to an additional infusion, an additional round of the LD chemotherapy regimen was permitted at the investigator's discretion based on patient's blood routine examination and if greater than 2 weeks since last infusion.

Statistical analysis

Comparisons of two groups were performed by the Mann–Whitney *U* test (Wilcoxon Rank Sum test), where $P < 0.05$ with a 95% confidence interval (CI) was considered statistically significant. For all clinical, laboratory, and cytokine markers, values less than the limit of detection were recorded as half the lower limit. Spearman rank coefficient test was used to perform all nonparametric correlations. GraphPad Prism 9 was used for statistical analysis.

Results

Patients

Between November 23, 2018, and the data cutoff on April 27, 2021, 16 patients were enrolled, and a total of 12 patients received ET019003 T cells. Four (4) patients did not receive an infusion because of an inability to manufacture as a result of poor T-cell activation (1 patient), high tumor burden (1 patient), or active infection (2 patients). As of the data cutoff date, the median duration of follow-up was 128 days (range: 34–728 days).

At enrollment, patients who received ET019003 T-cell therapy had a median age of 52 years (range: 36–79 years), a median of 3 previous therapies (range: 1–6), and a median ECOG score of 1 (range: 0–2). In general, ET019003-treated patients had high-risk features at baseline: 11 patients (92%) had stage III disease or higher based on the Ann Arbor staging system (McCarten et al. 2019), 9 patients (75%) had an international prognostic index (IPI) score of 3 or higher, and 8 patients (67%) had more than one extranodal involvement site. One individual (patient BH05-P06) previously received anti-CD19 T-cell therapy. No patients had previously undergone stem cell transplantation. Characteristics of the patients who received ET019003 T cells are shown in Table 1 and Online Supplemental Tables S1 and S2.

Dosing and lymphodepletion

Enrolled patients received a median of two (2) infusions of ET019003 T cells (range: 1–4 infusions). Characteristics of manufactured ET019003 T cells are provided in Online Supplemental Table S3. Following the first infusion, 9 of 12 patients (75%) received one or more additional infusions. The median time between infusions was 36 days (range: 17–73 days). Overall, the median total dose of ET019003 Receptor-positive T cells was 3.42×10^8 (range: 2.28×10^8 to 5.85×10^8 cells). Six of the 9 patients (67%) who received one or more additional infusions of ET019003 T cells also received a second round of the LD chemotherapy regimen. The median start date of the second round of LD chemotherapy was day 49 (range: day 43–day 85). Dosing details for each infused patient are shown in Online Supplemental Table S4.

Anti-tumor activity

Most patients who received ET019003 T-cell therapy responded to the treatment (Fig. 1a). The best ORR was 83% (95% CI 52–98%; Fig. 1b). Among the responders, 60% (6 of 10 responders) had a CR, including two individuals

Table 1 Baseline characteristics of patients, clinical response, and adverse events after ET019003 T-cell therapy

Patient number	Gender	Age (years)	Lymphoma type	Extranodal involvement	Disease stage	IPI Score	Refractory/relapsed status	ECOG	Previous lines of therapies ^a	Total dose ($\times 10^6$) ^b	CRS grade	Neurological toxicity grade	Best response ^c	Duration of response (days) ^d
BH05-P06	M	61	Non-GCB DLBCL	Left ¼ intercostal muscle, visceral pleura	III-E	3	Refractory	1	3	585	1	None	CR	51
BH05-P07	F	42	Non-GCB DLBCL	Left adrenal gland, right back subcutaneous tissue	III-A	3	Relapsed	0	4	486	2	None	CR	689+
BH05-P10	M	66	GCB DLBCL	None	IV	3	Refractory	2	5	275	1	None	CR	172+
BH05-P13	M	56	Non-GCB DLBCL	Gastric body, right adrenal gland, lumbar subcutaneous tissue	IV	3	Relapsed	1	3	342	1	None	CR	706+
BH05-P14	M	52	GCB DLBCL	Prostate gland, bladder	IV-A	4	Refractory	2	1	444	2	None	PR	24
BH05-P15	F	42	Non-GCB DLBCL	Breast, thyroid, pancreas, cervix uterus, lung, left temporal lobe, bones, subcutaneous tissue of head	IV-A	3	Relapsed	2	3	576	1	3	PR	59
BH05-P16	F	45	Non-GCB DLBCL	None	III-B	1	Relapsed	1	2	293	2	None	PR	35
BH05-P17	F	36	Non-GCB DLBCL	Gastric body, spleen	IV	3	Relapsed	1	6	342	none	None	SD	–

Table 1 (continued)

Patient number	Gender	Age (years)	Lymphoma type	Extranodal involvement	Disease stage	IPI Score	Refractory/relapsed status	ECOG	Previous lines of therapies ^a	Total dose ($\times 10^6$) ^b	CRS grade	Neurological toxicity grade	Best response ^c	Duration of response (days) ^d
BH05-P19	M	59	GCB DLBCL	Right fossa orbitalis, soft tissue next to the left rib	IV-A	2	Relapsed	1	2	555	1	None	PR	55
BH05-P20	M	79	Non-GCB DLBCL	Multiple nodules in both lungs (left upper lobe, right upper lobe, right middle lobe)	IV-B	3	Refractory	2	3	228	none	None	PD	–
BH05-P21	M	51	GCB DLBCL	Diaphragm near the right edge of the heart and the upper edge of the liver, right pleura, left adrenal gland	IV-A	3	Relapsed	1	5	228	1	None	CR	152
BH05-P22	M	56	FL (grade 3B)	None	II-A	0	Relapsed	0	2	255	1	None	CR	316+

CR complete response, CRS cytokine release syndrome, ECOG Eastern Cooperative Oncology Group, F female, FL follicular lymphoma, GCB DLBCL germinal center B-cell like diffuse large B-cell lymphoma, IPI score International Prognostic Index score, M male, PD progressive disease, PR partial response, SD stable disease, + Ongoing response

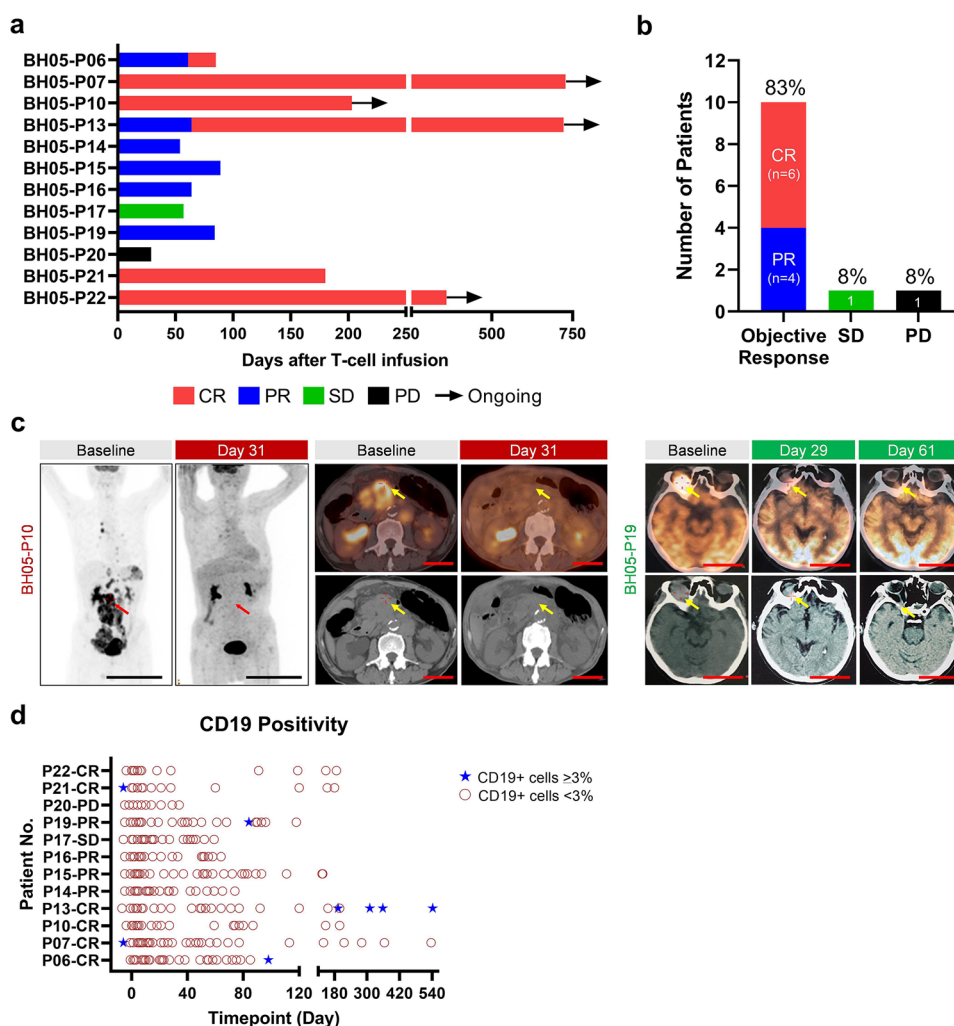
^aList of previous therapies for each patient is provided in Online Supplemental Table S1

^bDose is given as number of ET019003 Receptor-positive T cells

^cBest response was defined as the best response that a patient achieved after ET019003 T-cell infusion

^dDuration of response (DoR) was defined as the time from the first recorded response (CR or PR) to the date of first recorded progression or death due to B-cell lymphoma. If a patient had not progressed or died at the time of data cutoff, DoR was censored on the date for the last available follow-up visit

Fig. 1 Clinical responses to ET019003 T-cell therapy. **a** Treatment response and duration of response after initial infusion of ET019003 T cells. Black arrows indicate ongoing remission and follow-up. **b** Best response for the twelve patients. Best response was defined as the best response (i.e., CR > PR > SD > PD) the patient achieved at any time after ET019003 T-cell therapy. CR complete response, PR partial response, SD stable disease, PD progressive disease. **c** Representative radiographic images of two responders (BH05-P10 and BH05-P19) at baseline and the indicated time points after ET019003 T-cell therapy. Red or yellow arrows mark the tumor lesions. Full body images are PET-CT scans. Cross-sectional images are PET scans (top rows) and CT scans (bottom rows). Scale bars: black, 20 cm; red, 6 cm. **d** Percentage (%) of CD19-positive lymphocytes in peripheral blood samples by means of flow cytometry. Negative results were defined as less than 3% of lymphocytes that were positive for CD19



who initially had a PR that improved to CR, and 40% (4 of 10 responders) had a PR. Representative PET-CT scans of four (4) of the 10 patients who achieved either a CR or PR are shown in Fig. 1c and Online Supplemental Figure S2. Only two (2) patients did not respond to ET019003 T-cell therapy, including one individual (patient BH05-P17) who at the Month 1 assessment had stable disease (SD) that subsequently progressed when evaluated at the Month 2 assessment. Importantly, among the patients who achieved a CR, 67% (4 of 6 patients) remain in ongoing CR as of the patient's last scan on or before the data cut-off date; this includes 2 patients with ongoing CRs for 22.7 and 23.2 months. As of the data cut-off date, the median DoR for the six patients who achieved CR was > 244 days (range: 51–706 days).

B-cell aplasia

B-cell aplasia (absence of CD19-positive cells) occurred in all patients and 10 of 12 patients (83%) had B-cell aplasia

at last follow-up (Fig. 1d and Online Supplemental Figure S3). Reductions in immunoglobulins (IgA, IgG, and IgM) were also observed in some of the treated patients (Online Supplemental Figures S4–S6). Most patients (10 of 12; 83%) required immunoglobulin replacement treatment following ET019003 T-cell therapy.

ET019003 T-cell expansion and persistence

ET019003 T-cell expansion and persistence were assessed in the peripheral blood (PB) after ET019003 T-cell therapy by ET019003 transgene-specific quantitative polymerase chain reaction (qPCR) and by flow cytometry using an anti-idiotypic antibody directed against ET190L1 (Xu et al. 2018). The levels of ET019003 T cells as detected by qPCR correlated well with the level of ET019003 T cells as detected by flow cytometry (Spearman $r=0.85$, $P=0.0008$; Online Supplemental Figure S7A).

ET019003 T-cell expansion was seen in all treated patients (Fig. 2 and Online Supplemental Figure S8). In the

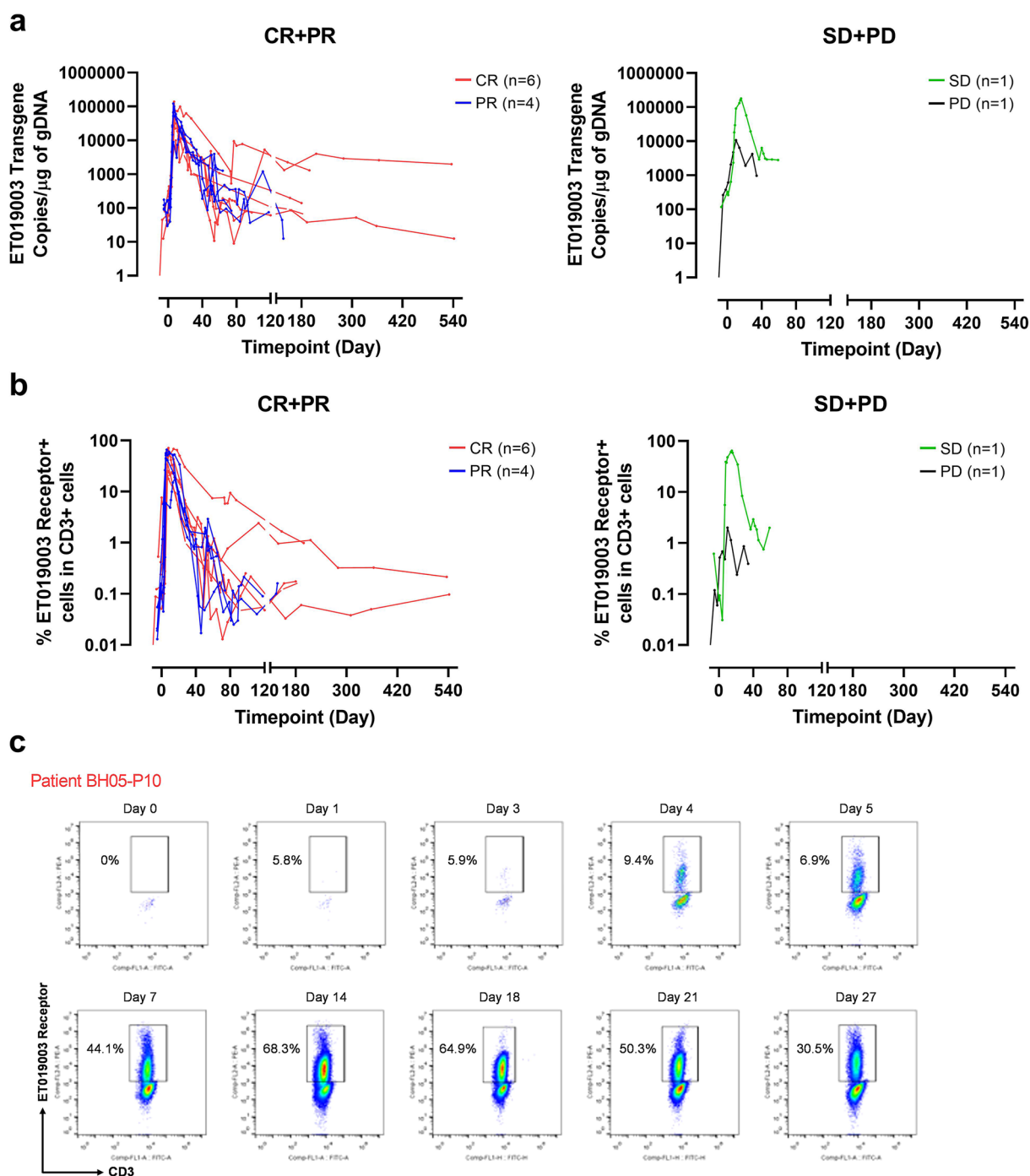


Fig. 2 Kinetics of T-cell expansion after ET019003 T-cell infusion. ET019003 T-cell expansion and persistence in peripheral blood was assessed by **a** qPCR and **b** flow cytometry. Patients' best response are denoted by colors: CR (red), PR (blue), SD (green), and PD (black). **c** Flow cytometry analysis showing in vivo expansion of ET019003

Receptor-positive T cells in the peripheral blood of representative patient BH05-P10, who had a CR. For the qPCR assay, values less than the limit of detection (25 copies/ μ g of gDNA) were recorded as half the lower limit

10 patients who had a response (CR or PR), the median peak concentration (C_{max}) of ET019003 T cells in the PB was 63,000 copies/ μ g of genomic DNA (gDNA; range: 11,700 to 139,000) (Fig. 2a). Interestingly, one of the two

non-responders (patient BH05-P17) had the highest C_{max} with 180,000 copies/ μ g of gDNA, while the other non-responder (patient BH05-P20) had the lowest C_{max} with 10,700 copies/ μ g of gDNA.

While 9 of 12 patients received at least a second infusion of ET019003 T cells, all 12 patients reached C_{max} prior to administration of a second dose (if given). The median time of peak concentration (T_{max}) of ET019003 T cells in PB was day 8 (range: day 6–day 16). Further T-cell expansion, defined as greater than twofold increase in ET019003 T-cell levels (copies/μg of gDNA) from immediately before (i.e., within 24 h of infusion) to after infusion, was observed in 2 of 9 patients (22%) who received a second infusion and in 3 of 4 patients (75%) who received a third infusion. Further T-cell expansion was not observed in patient BH05-P06 following the fourth infusion. Of note, further T-cell expansion was more likely to occur and to a higher degree in re-treated patients if LD chemotherapy was administered prior to the infusion (Online Supplemental Figure S9). Interestingly, the degree of expansion (i.e., median C_{max}) was modestly higher in patients with grade 2 CRS ($n=3$; median C_{max}: 101,000 copies/μg of gDNA) compared to patients with grade 0–1 CRS ($n=9$; median C_{max}: 59,800 copies/μg of gDNA) however, the difference was not statistically significant ($P=0.6000$; Online Supplemental Figure S8A).

In responders ($n=10$ patients), the median peak proportions of ET019003 Receptor-positive cells in CD3-positive cells was 49.2% (range: 18.6 to 71.3) (Fig. 2b, c). One of the two non-responders (patient BH05-P17) had a peak level of 64.4%, while the other non-responder (patient BH05-P20) had a peak level of 2.0%.

The longest follow-up that included pharmacokinetic (PK) assessment was 542 days (median: 128 days; range: 34–542). Importantly, ET019003 T cells continued to be detected (defined as $>0.1\%$ ET019003 Receptor-positive cells of total CD3-positive cells) in 10 of 12 (83%) patients at last follow-up.

Safety

Adverse events classified as possibly, probably, or definitely related to ET019003 T-cell therapy are shown in Table 1 and Online Supplemental Tables S5–S12. Overall, ET019003 T-cell therapy was well-tolerated. No severe CRS (grade 3 or higher) was observed in any of the 12 patients. Mild CRS (grade 1 or 2) was observed after initial T-cell infusion in all 10 patients who had a response (CR or PR). Of the 10 responders, 7 patients experienced grade 1 CRS and 3 patients experienced grade 2 CRS. In all cases, CRS was only observed following first T-cell infusion. No CRS was observed in patients after a second, third, or fourth T-cell infusion. Importantly, no patients required tocilizumab treatment. No CRS (of any grade) was observed in non-responders ($n=2$). No neurological toxicities were observed in 11 of 12 patients (92%). One patient (BH05-P15), who prior to ET019003 T-cell therapy had lymphoma invasion in the central nervous system, experienced grade 3 neurotoxicity

after T-cell infusion. This patient was treated with glucocorticoid and the symptoms resolved within 24 h. As expected, hematologic toxicities (i.e., bone marrow suppression), including neutropenia, thrombocytopenia, and anemia occurred in most patients and were attributed to the pre-infusion LD chemotherapy regimen (Online Supplemental Figures S11–S14).

Release of cytokines and inflammatory markers

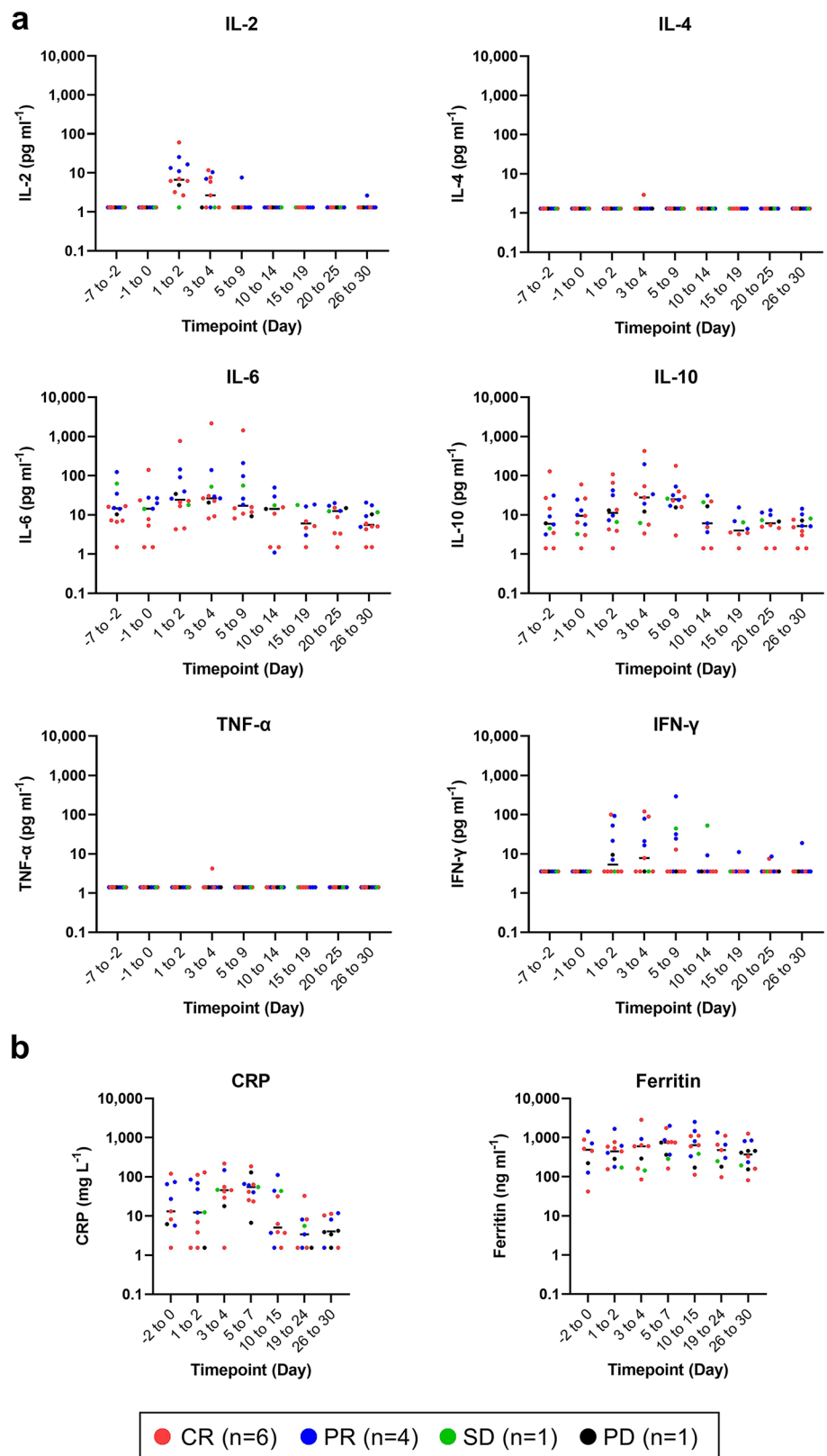
CAR T-cell therapy is routinely associated with elevated levels of cytokines such as interleukin (IL)-2, IL-6, tumor necrosis factor- α (TNF- α), and interferon- γ (IFN- γ), as well as elevated levels of serum inflammatory markers such as C-reactive protein (CRP) and ferritin (Grupp et al. 2013; Porter et al. 2015; Teachey et al. 2016; Turtle et al. 2016a). The highest peak concentrations of these cytokines and inflammatory markers are seen in patients with severe CRS and/or neurotoxicity (Teachey et al. 2016; Turtle et al. 2016a). Following initial ET019003 infusion, a panel of CRS-related cytokines (IL-2, IL-4, IL-6, IL-10, TNF- α , and IFN- γ) and inflammatory markers (CRP and ferritin) were serially measured in patient samples.

During the first 30 days after ET019003 infusion, we observed no meaningful elevation in levels of IL-4 or TNF- α in any patient (Fig. 3a). In general, levels of both IL-4 and TNF- α remained at or below the limit of detection (median IL-4 peak: 1.3 pg/ml, range: 1.3–2.9, and median TNF- α peak: 1.4 pg/ml, range: 1.4–4.3). Likewise, most patients (11 of 12; 92%) experienced increases in IL-2, IL-6, IL-10, and IFN- γ that peaked within 10 days after infusion (Fig. 3a). Median peak levels for these patients were as follows: IL-2 (7.1 pg/ml, range: 1.3–25.4); IL-6 (34.4 pg/ml, range: 16.7–211); IL-10 (27.8 pg/ml, range: 3.4–197); and IFN- γ (21.7 pg/ml, range: 3.6–296). In contrast to most patients, one subject (patient BH05-P10) showed larger increases in several cytokines, including IL-2 (peak value: 60.5 pg/ml), IL-10 (peak value: 429 pg/ml), IL-6 (peak value: 2,170 pg/ml), and IFN- γ (peak value: 120 pg/ml).

Prior to initial ET019003 infusion, baseline ferritins ($n=8$) were elevated in most patients (median: 488 ng/ml, range: 42.1–1430). Only 3 patients out of the 8 measured had baseline ferritins <400 ng/ml. After initial ET019003 infusion, ferritins increased on average by ~ 2.8 -fold, peaking on average on day 8 (median peak: 767 ng/ml, range: 162–2870; Fig. 3b). Baseline CRP ($n=9$) was also elevated in most patients (median: 13.1 mg/L, range: 1.6–120). Only four patients out of the nine measured had a baseline CRP <10 mg/L. After initial ET019003 infusion, CRP levels modestly increased in all patients, peaking on average on day 4 (median peak: 64.7 mg/L, range: 3.8 to 219; Fig. 3b).

Peak CRP levels correlated with peak ferritin (Spearman $r=0.74$, $P=0.0078$) and peak IL-6 levels (Spearman

Fig. 3 Levels of cytokines and serum inflammatory markers after ET019003 T-cell infusion. **a** Cytokine levels and **b** Serum CRP and ferritin levels in patients during the first month of ET019003 T-cell therapy. Horizontal lines denote median values. Patients' best response are denoted by color of the symbols: CR (red), PR (blue), SD (green), and PD (black). Values less than the limit of detection (LOD; see Online Supplemental data for LOD values) were recorded as half the lower limit



$r=0.63$, $P=0.0323$; Online Supplemental Figures S7B and S7C). However, there was no correlation between peak CRP, ferritin, and IL-6 levels with the peak concentration (C_{max}) of ET019003 T cells in the patient's blood after infusion (Online Supplemental Figures S7D–S7F). The median peak CRP value, ferritin value, and IL-6 value were all slightly higher in patients with grade 2 CRS ($n=3$) compared to patients with grade 0–1 CRS ($n=9$). However, the median differences were not statistically significant ($P=0.7273$, $P=0.6303$, and $P=0.8636$, respectively; Online Supplemental Figures S8B–S8D). In addition to levels of CRP and ferritin, levels of other blood proteins (e.g., ALT, AST, and LDH) were also examined and deemed not clinically significant (Online Supplemental Figures S15–S21).

Discussion

CAR T-cell therapies have demonstrated a clear survival benefit in individuals with refractory hematological malignancies (Neelapu et al. 2017; Maude et al. 2018; Raje et al. 2019; Wang et al. 2020; Munshi et al. 2021). However, the efficacy of CAR-based therapies comes at the expense of high rates of CRS and ICANS (Pennisi et al. 2020), which underscores the need for novel approaches to the CAR design that can reduce toxicities. To this end, we recently developed the AbTCR platform. The AbTCR construct links antibody-based target recognition to a human $\gamma\delta$ TCR (Xu et al. 2018), allowing natural formation of a multimeric receptor with the endogenous CD3 complex and engaging pathways of activation and regulation normally employed by endogenous TCRs. In addition, unlike CARs, which include an *in-cis*, constitutive, co-stimulatory domain that is activated whenever the CARs bind to their target, the AbTCR construct does not have such a co-stimulatory domain. Instead, AbTCR T cells are engineered to co-express AbTCR and a separate co-stimulatory molecule that is customized to recognize a tumor-specific antigen. These features were designed into the AbTCR platform to enable anti-tumor activity similar to CAR T cells, but with the added benefit of less T-cell mediated toxicity.

ET019003 T-cell therapy is built on our novel technology platform and includes an AbTCR construct and a co-stimulatory molecule, both designed to recognize and bind the CD19 antigen. Here, we report the results of our early phase I clinical study of ET019003 T cells for the treatment of CD19-positive r/r B-cell lymphoma. In a cohort of high-risk patients who had undergone a median of three previous therapies, ET019003 T-cell therapy resulted in a best ORR of 83% (95% CI 52–98), of which 6 of 12 patients (50%, 95% CI 21–79) achieved a CR, and 4 of 12 patients (33%, 95% CI 10–65) achieved a PR. These response rates are at least comparable to those reported in published studies with

anti-CD19 CAR T-cell therapies in adult B-cell lymphoma (Kochenderfer et al. 2017; Neelapu et al. 2017; Locke et al. 2019; Schuster et al. 2019; Ying et al. 2019). While two infused patients achieved a PR that subsequently converted to a CR, the remaining PRs were not sustainable, with a median DoR of 45 days (range: 24–59 days). As of the data cut-off date, patients who either never responded ($n=2$) or responded yet subsequently progressed ($n=6$) either had succumbed to their disease ($n=5$) or pursued other treatment options for their disease ($n=3$). Importantly, clinical responses in patients achieving CR were durable. As of the patients' last follow-up visit on or before the data cut-off date, 4 of 6 patients (67%) who attained a CR remain with ongoing CR. The median time these four patients have been in remission is 16.6 months (range: 5.7–23.2 months), including two patients who have been in complete remission for 22.7 and 23.2 months.

Pronounced ET019003 T-cell expansion was observed in all infused patients, albeit to a lesser extent in one of the two non-responders. At maximal expansion, ET019003 T cells comprised up to ~71% of circulating T cells with a geometric mean C_{max} of 52,900 copies/ μ g of gDNA, which was achieved in all patients shortly after infusion (T_{max} : day 8). Importantly, 3 of the 4 patients with ongoing CR had ongoing persistence of ET019003 T cells with associated B-cell aplasia, including one patient with detectable levels of circulating ET019003 T cells at 17.6 months post-infusion. Overall, these data demonstrate the robust proliferative potential and persistence of ET019003 T cells *in vivo*.

ET019003 T-cell therapy had an attractive safety profile consistent with the intended design of the AbTCR platform. There was no severe CRS (grade ≥ 3) and only one case of neurotoxicity (of any grade) which developed in a patient with widespread extranodal disease, including nodules in the brain. This patient experienced grade 3 neurotoxicity 4 days after infusion that manifested primarily as cognitive disturbance and that completely resolved within 24 h following glucocorticoid treatment. Although we cannot rule out differences due to small sample size and patient characteristics, these preliminary findings are in contrast to what is typically observed with CD19 CAR T-cell therapies. For example, among seven (7) clinical trials with CAR T cells directed against non-Hodgkin lymphoma, one study reported rates of ICANS (of any grade) as high as 65%, while another study reported rates of severe ICANS (grade ≥ 3) as high as 55% (Turtle et al. 2016b; Kochenderfer et al. 2017; Neelapu et al. 2017; Schuster et al. 2017; Jaglowski et al. 2019; Pasquini et al. 2019; Brudno et al. 2020). According to the Center for International Blood and Marrow Transplant Research (CIBMTR) registry, however, rates of CRS and ICANS appear lower in the real-world setting compared to pivotal trials, at least among patients who received commercial tisagenlecleucel (KymriahTM) CAR-T therapy (Pasquini et al. 2020).

Nevertheless, CRS and ICANS continue to be of concern and, although extremely rare, can lead to death. Moreover, looking beyond hematological malignancies, recent solid tumor studies have reported patient deaths attributed to CAR T-related toxicity, including macrophage activation syndrome (Carroll 2021), which further highlights the need for safer strategies regardless of cancer type.

In addition, infusion of ET019003 T cells also did not result in any significant elevation of cytokine levels in treated patients, even with robust ET019003 T-cell expansion. Peak cytokine levels in all patients were significantly lower than previously reported, including one study that reported IL-6 levels as high as 20,892 pg/ml (median: 122 pg/ml, range: 0.40–20,892) in a cohort of leukemia patients ($n=37$) with grade 0–3 CRS following CTL019 CAR T-cell treatment and as high as 102,476 pg/ml (median: 8309 pg/ml, range: 580–102,476) in a cohort of patients ($n=14$) with grade 4–5 CRS (Teachey et al. 2016). Peak ferritin and CRP levels were also lower than typically observed following CD19 CAR T-cell therapy, including one study that reported a median peak ferritin value of 23,439 ng/ml (range: 280–411,936) and a median peak CRP value of 201 mg/L (range: 7–565) in leukemia patients ($n=51$) treated with CTL019 CAR T cells (Teachey et al. 2016). Likewise, a separate study reported a median peak CRP value of ~950 mg/L (~15 times higher than what was observed following ET019003 treatment) in lymphoma patients ($n=16$) who responded after CAR-19 T-cell therapy (Kochenderfer et al. 2017). Overall, these data are in concordance with our preclinical data that showed AbTCR-mediated T-cell activation results in low cytokine release both in vitro and in mouse models of leukemia and lymphoma (Xu et al. 2018).

In summary, this early phase I clinical study demonstrates that ET019003 T-cell therapy has anti-tumor activity and an attractive safety profile in patients with CD19-positive r/r B-cell lymphoma. While additional studies are required to confirm the results of this small study, the findings are consistent with the design of the AbTCR platform as a safer but equally effective alternative to other engineered T-cell therapies such as CAR T-cell therapies. Additional therapies based on the AbTCR platform are currently being studied in a variety of tumor types.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00432-022-04132-9>.

Acknowledgements The authors thank Jing Li, Xin Liu, Mengchang Wang, Jieying Xi, and Haitao Zhang for their involvement in the conduct of the study; Hong Liu for assistance with clinical protocol design; Joy Lin for help with data organization and manuscript preparation; and Vivien Chan, Sal Fuerstenberg, Mei Hong, Michael Kavanaugh, Nicole Nuñez, and Pei Wang for critical review of the manuscript.

Author contributions PH, MZ, HL, and CL conceived and designed the study; PH oversaw the conduct of the study; HL, JW, ML, HL, LC,

and HW were involved in the conduct of the study; HL, BZ, and FT analyzed and interpreted clinical data; BZ analyzed and interpreted correlative data and performed statistical analysis; QC oversaw drug product manufacturing, Quality Control, and correlative studies; FT collected correlative data; FT and DY collected drug product data; FN oversaw the Good Manufacturing Practice laboratory and drug product manufacturing; DY analyzed drug product data and established Quality Control testing for drug product release; SAG assisted with the interpretation of clinical and correlative data; BZ wrote the manuscript; and all authors critically reviewed the manuscript and approved the content.

Funding This study was sponsored by The First Affiliated Hospital of Xi'an Jiaotong University in collaboration with Eureka Therapeutics, Inc. This study was supported by National Key R & D Program of China (No. 2019YFC1316204), Key Research and Development Program of Shaanxi (No. 2018ZDCXL-SF-01–02-01, 2021ZDLSF02-17 and 2021SF-302), and National Scientific and Technological Major Special Project for Significant Creation of New Drugs (No. 2020ZX09201020).

Data availability All data generated or analyzed during this study are included in this published article and its supplementary information file.

Declarations

Competing interests BZ, QC, FT, FN, DY, and CL are employed at Eureka Therapeutics, Inc., (Eureka) and have equity and/or stock options in Eureka. CL serves on advisory boards for the following companies: Eureka and JW Therapeutics. SAG has received support from Novartis, Servier, and Kite and serves as a consultant, member of the scientific advisory board or study steering committee for Novartis, Cellectis, Adaptimmune, Eureka, TCR2, Juno, GlaxoSmithKline, Vertex, Cure Genetics, Humanigen, and Roche. The remaining authors declare no competing financial interests.

Ethical approval This study was approved by the Ethics Committee of The First Affiliated Hospital of Xi'an Jiaotong University in China. The study was conducted in accordance with the principles of the Declaration of Helsinki.

Consent to participate All participants in the study provided written informed consent.

Consent to publish Patient consent for publication is not required. Patients consented to participate in the study.

References

- Abramson JS, Palomba ML, Gordon LI, Lunning MA, Wang M, Arnaason J, Mehta A, Purev E, Maloney DG, Andreadis C, Sehgal A, Solomon SR, Ghosh N, Albertson TM, Garcia J, Kostic A, Mallaney M, Ogasawara K, Newhall K, Kim Y, Li D, Siddiqi T (2020) Lisocabtagene maraleucel for patients with relapsed or refractory large B-cell lymphomas (TRANSCEND NHL 001): a multicentre seamless design study. *Lancet* 396(10254):839–852. [https://doi.org/10.1016/s0140-6736\(20\)31366-0](https://doi.org/10.1016/s0140-6736(20)31366-0)
- Barrett DM, Singh N, Porter DL, Grupp SA, June CH (2014) Chimeric antigen receptor therapy for cancer. *Annu Rev Med* 65:333–347. <https://doi.org/10.1146/annurev-med-060512-150254>
- Brentjens RJ, Santos E, Nikhamin Y, Yeh R, Matsushita M, La Perle K, Quintás-Cardama A, Larson SM, Sadelain M (2007) Genetically

- targeted T cells eradicate systemic acute lymphoblastic leukemia xenografts. *Clin Cancer Res* : off J Am Assoc Cancer Res 13(18 Pt 1):5426–5435. <https://doi.org/10.1158/1078-0432.ccr-07-0674>
- Brudno JN, Lam N, Vanasse D, Shen YW, Rose JJ, Rossi J, Xue A, Bot A, Scholler N, Mikkilineni L, Roschewski M, Dean R, Cachau R, Youkharibache P, Patel R, Hansen B, Stroncek DF, Rosenberg SA, Gress RE, Kochenderfer JN (2020) Safety and feasibility of anti-CD19 CAR T cells with fully human binding domains in patients with B-cell lymphoma. *Nat Med* 26(2):270–280. <https://doi.org/10.1038/s41591-019-0737-3>
- Carroll J (2021) Carl June's Tmunity encounters a lethal roadblock as 2 patient deaths derail lead trial, raise red flag forcing a rethink of CAR-T for solid tumors. *Endpoint News*. <https://endpts.com/exclusive-carl-junes-tmunity-encounters-a-lethal-roadblock-as-2-patient-deaths-derail-lead-trial-raise-red-flag-forcing-a-rethink-of-car-t-for-solid-tumors/>. Accessed June 25, 2021
- Chen L, Flies DB (2013) Molecular mechanisms of T cell co-stimulation and co-inhibition. *Nat Rev Immunol* 13(4):227–242. <https://doi.org/10.1038/nri3405>
- Cheson BD, Fisher RI, Barrington SF, Cavalli F, Schwartz LH, Zucca E, Lister TA (2014) Recommendations for initial evaluation, staging, and response assessment of Hodgkin and non-Hodgkin lymphoma: the Lugano classification. *J Clin Oncol* 32(27):3059–3068. <https://doi.org/10.1200/jco.2013.54.8800>
- Gauthier J, Turtle CJ (2018) Insights into cytokine release syndrome and neurotoxicity after CD19-specific CAR-T cell therapy. *Curr Res Transl Med* 66(2):50–52. <https://doi.org/10.1016/j.retram.2018.03.003>
- Grupp SA, Kalos M, Barrett D, Aplenc R, Porter DL, Rheingold SR, Teachey DT, Chew A, Hauck B, Wright JF, Milone MC, Levine BL, June CH (2013) Chimeric antigen receptor-modified T cells for acute lymphoid leukemia. *N Engl J Med* 368(16):1509–1518. <https://doi.org/10.1056/NEJMoa1215134>
- Jagłowski S, Hu Z-H, Zhang Y, Kamdar M, Ghosh M, Lulla P, Sasine J, Perales M-A, Hematti P, Nikiforow S, Steinert P, Jeschke M, Yi L, Chawla R, Pacaud L, Horowitz MM, Bleickardt E, Pasquini MC (2019) Tisagenlecleucel chimeric antigen receptor (CAR) T-cell therapy for adults with diffuse large B-cell lymphoma (DLBCL): real world experience from the center for international blood and marrow transplant research (CIBMTR) cellular therapy (CT) registry. *Blood* 134(Supplement_1):766. <https://doi.org/10.1182/blood-2019-130983>
- Kochenderfer JN, Somerville RPT, Lu T, Shi V, Bot A, Rossi J, Xue A, Goff SL, Yang JC, Sherry RM, Klebanoff CA, Kammala US, Sherman M, Perez A, Yuan CM, Feldman T, Friedberg JW, Roschewski MJ, Feldman SA, McIntyre L, Toomey MA, Rosenberg SA (2017) Lymphoma remissions caused by anti-CD19 chimeric antigen receptor T cells are associated with high serum interleukin-15 levels. *J Clin Oncol* 35(16):1803–1813. <https://doi.org/10.1200/jco.2016.71.3024>
- Lee DW, Santomaso BD, Locke FL, Ghobadi A, Turtle CJ, Brudno JN, Maus MV, Park JH, Mead E, Pavletic S, Go WY, Eldjerou L, Gardner RA, Frey N, Curran KJ, Peggs K, Pasquini M, DiPersio JF, van den Brink MRM, Komanduri KV, Grupp SA, Neelapu SS (2019) ASTCT consensus grading for cytokine release syndrome and neurologic toxicity associated with immune effector cells. *Biol Blood Marrow Transplant* : J Am Soc Blood Marrow Transplant 25(4):625–638. <https://doi.org/10.1016/j.bbmt.2018.12.758>
- Locke FL, Ghobadi A, Jacobson CA, Miklos DB, Lekakis LJ, Oluwole OO, Lin Y, Braunschweig I, Hill BT, Timmerman JM, Deol A, Reagan PM, Stiff P, Flinn IW, Farooq U, Goy A, McSweeney PA, Munoz J, Siddiqi T, Chavez JC, Herrera AF, Bartlett NL, Wiecek JS, Navale L, Xue A, Jiang Y, Bot A, Rossi JM, Kim JJ, Go WY, Neelapu SS (2019) Long-term safety and activity of axicabtagene ciloleucel in refractory large B-cell lymphoma (ZUMA-1): a single-arm, multicentre, phase 1–2 trial. *Lancet Oncol* 20(1):31–42. [https://doi.org/10.1016/s1470-2045\(18\)30864-7](https://doi.org/10.1016/s1470-2045(18)30864-7)
- Maude SL, Barrett D, Teachey DT, Grupp SA (2014) Managing cytokine release syndrome associated with novel T cell-engaging therapies. *Cancer J* 20(2):119–122. <https://doi.org/10.1097/ppo.0000000000000035>
- Maude SL, Teachey DT, Porter DL, Grupp SA (2015) CD19-targeted chimeric antigen receptor T-cell therapy for acute lymphoblastic leukemia. *Blood* 125(26):4017–4023. <https://doi.org/10.1182/blood-2014-12-580068>
- Maude SL, Laetsch TW, Buechner J, Rives S, Boyer M, Bittencourt H, Bader P, Verneris MR, Stefanski HE, Myers GD, Qayed M, De Moerloose B, Hiramatsu H, Schlis K, Davis KL, Martin PL, Nemecek ER, Yanik GA, Peters C, Baruchel A, Boissel N, Mechinaud F, Balduzzi A, Krueger J, June CH, Levine BL, Wood P, Taran T, Leung M, Mueller KT, Zhang Y, Sen K, Lebwohl D, Pulsipher MA, Grupp SA (2018) Tisagenlecleucel in children and young adults with B-cell lymphoblastic leukemia. *N Engl J Med* 378(5):439–448. <https://doi.org/10.1056/NEJMoa1709866>
- McCarten KM, Nadel HR, Shulkin BL, Cho SY (2019) Imaging for diagnosis, staging and response assessment of Hodgkin lymphoma and non-Hodgkin lymphoma. *Pediatr Radiol* 49(11):1545–1564. <https://doi.org/10.1007/s00247-019-04529-8>
- Munshi NC, Anderson LD Jr, Shah N, Madduri D, Berdeja J, Lonial S, Raje N, Lin Y, Siegel D, Oriol A, Moreau P, Yakoub-Agha I, Delforge M, Cavo M, Einsele H, Goldschmidt H, Weisel K, Rambaldi A, Reece D, Petrocchi F, Massaro M, Connarn JN, Kaiser S, Patel P, Huang L, Campbell TB, Hege K, San-Miguel J (2021) Idecabtagene vicleucel in relapsed and refractory multiple myeloma. *N Engl J Med* 384(8):705–716. <https://doi.org/10.1056/NEJMoa2024850>
- Neelapu SS, Locke FL, Bartlett NL, Lekakis LJ, Miklos DB, Jacobson CA, Braunschweig I, Oluwole OO, Siddiqi T, Lin Y, Timmerman JM, Stiff PJ, Friedberg JW, Flinn IW, Goy A, Hill BT, Smith MR, Deol A, Farooq U, McSweeney P, Munoz J, Avivi I, Castro JE, Westin JR, Chavez JC, Ghobadi A, Komanduri KV, Levy R, Jacobsen ED, Witzig TE, Reagan P, Bot A, Rossi J, Navale L, Jiang Y, Aycock J, Elias M, Chang D, Wiecek J, Go WY (2017) Axicabtagene ciloleucel CAR T-cell therapy in refractory large B-cell lymphoma. *N Engl J Med* 377(26):2531–2544. <https://doi.org/10.1056/NEJMoa1707447>
- Park JH, Rivière I, Gonen M, Wang X, Sénéchal B, Curran KJ, Sauter C, Wang Y, Santomaso B, Mead E, Roshal M, Maslak P, Davila M, Brentjens RJ, Sadelain M (2018) Long-term follow-up of CD19 CAR therapy in acute lymphoblastic leukemia. *N Engl J Med* 378(5):449–459. <https://doi.org/10.1056/NEJMoa1709919>
- Pasquini MC, Hu Z-H, Curran K, Laetsch T, Locke F, Rouce R, Pulsipher MA, Phillips CL, Keating A, Frigault MJ, Salzberg D, Jagłowski S, Sasine JP, Rosenthal J, Ghosh M, Landsburg D, Margossian S, Martin PL, Kamdar MK, Hematti P, Nikiforow S, Turtle C, Perales M-A, Steinert P, Horowitz MM, Moskop A, Pacaud L, Yi L, Chawla R, Bleickardt E, Grupp S (2020) Real-world evidence of tisagenlecleucel for pediatric acute lymphoblastic leukemia and non-Hodgkin lymphoma. *Blood Adv* 4(21):5414–5424. <https://doi.org/10.1182/bloodadvances.2020003092>
- Pasquini MC, Locke FL, Herrera AF, Siddiqi T, Ghobadi A, Komanduri KV, Hu Z-H, Dong H, Hematti P, Nikiforow S, Steinert P, Purdum A, Horowitz MM, Hooper M, Kawashima J, Jacobson CA (2019) Post-marketing use outcomes of an anti-CD19 chimeric antigen receptor (CAR) T cell therapy, axicabtagene ciloleucel (Axi-Cel), for the treatment of large B cell lymphoma (LBCL) in the United States (US). *Blood* 134(1):764. <https://doi.org/10.1182/blood-2019-124750>
- Pennisi M, Jain T, Santomaso BD, Mead E, Wudhikarn K, Silverberg ML, Batlevi Y, Shouval R, Devlin SM, Batlevi C, Brentjens RJ, Dahi PB, Diamante C, Giralt S, Halton EF, Maloy M, Palomba

- ML, Sanchez-Escamilla M, Sauter CS, Scordo M, Shah G, Park JH, Perales MA (2020) Comparing CAR T-cell toxicity grading systems: application of the ASTCT grading system and implications for management. *Blood Adv* 4(4):676–686. <https://doi.org/10.1182/bloodadvances.2019000952>
- Porter DL, Hwang WT, Frey NV, Lacey SF, Shaw PA, Loren AW, Bagg A, Marcucci KT, Shen A, Gonzalez V, Ambrose D, Grupp SA, Chew A, Zheng Z, Milone MC, Levine BL, Melenhorst JJ, June CH (2015) Chimeric antigen receptor T cells persist and induce sustained remissions in relapsed refractory chronic lymphocytic leukemia. *Sci Transl Med* 7(303):303ra139. <https://doi.org/10.1126/scitranslmed.aac5415>
- Raje N, Berdeja J, Lin Y, Siegel D, Jagannath S, Madduri D, Liedtke M, Rosenblatt J, Maus MV, Turka A, Lam LP, Morgan RA, Friedman K, Massaro M, Wang J, Russotti G, Yang Z, Campbell T, Hege K, Petrocca F, Quigley MT, Munshi N, Kochenderfer JN (2019) Anti-BCMA CAR T-cell therapy bb2121 in relapsed or refractory multiple myeloma. *N Engl J Med* 380(18):1726–1737. <https://doi.org/10.1056/NEJMoa1817226>
- Schuster SJ, Svoboda J, Chong EA, Nasta SD, Mato AR, Anak Ö, Brogdon JL, Pruteanu-Malinici I, Bhoj V, Landsburg D, Wasik M, Levine BL, Lacey SF, Melenhorst JJ, Porter DL, June CH (2017) Chimeric antigen receptor T cells in refractory B-cell lymphomas. *N Engl J Med* 377(26):2545–2554. <https://doi.org/10.1056/NEJMoa1708566>
- Schuster SJ, Bishop MR, Tam CS, Waller EK, Borchmann P, McGuirk JP, Jäger U, Jaglowski S, Andreadis C, Westin JR, Fleury I, Bachanova V, Foley SR, Ho PJ, Mielke S, Magenau JM, Holte H, Pantano S, Pacaud LB, Awasthi R, Chu J, Anak Ö, Salles G, Maziarsz RT (2019) Tisagenlecleucel in adult relapsed or refractory diffuse large B-cell lymphoma. *N Engl J Med* 380(1):45–56. <https://doi.org/10.1056/NEJMoa1804980>
- Teachey DT, Lacey SF, Shaw PA, Melenhorst JJ, Maude SL, Frey N, Pequignot E, Gonzalez VE, Chen F, Finklestein J, Barrett DM, Weiss SL, Fitzgerald JC, Berg RA, Aplenc R, Callahan C, Rheingold SR, Zheng Z, Rose-John S, White JC, Nazimuddin F, Wertheim G, Levine BL, June CH, Porter DL, Grupp SA (2016) Identification of predictive biomarkers for cytokine release syndrome after chimeric antigen receptor T-cell therapy for acute lymphoblastic leukemia. *Cancer Discov* 6(6):664–679. <https://doi.org/10.1158/2159-8290.cd-16-0040>
- Turtle CJ, Hanafi LA, Berger C, Gooley TA, Cherian S, Hudecek M, Sommermeyer D, Melville K, Pender B, Budiarto TM, Robinson E, Steevens NN, Chaney C, Soma L, Chen X, Yeung C, Wood B, Li D, Cao J, Heimfeld S, Jensen MC, Riddell SR, Maloney DG (2016a) CD19 CAR-T cells of defined CD4+:CD8+ composition in adult B cell ALL patients. *J Clin Invest* 126(6):2123–2138. <https://doi.org/10.1172/jci85309>
- Turtle CJ, Hanafi LA, Berger C, Hudecek M, Pender B, Robinson E, Hawkins R, Chaney C, Cherian S, Chen X, Soma L, Wood B, Li D, Heimfeld S, Riddell SR, Maloney DG (2016b) Immunotherapy of non-Hodgkin's lymphoma with a defined ratio of CD8+ and CD4+ CD19-specific chimeric antigen receptor-modified T cells. *Sci Transl Med* 8(355):355ra116. <https://doi.org/10.1126/scitranslmed.aaf8621>
- Wang M, Munoz J, Goy A, Locke FL, Jacobson CA, Hill BT, Timmerman JM, Holmes H, Jaglowski S, Flinn IW, McSweeney PA, Miklos DB, Pagel JM, Kersten MJ, Milpied N, Fung H, Topp MS, Houot R, Beitinjaneh A, Peng W, Zheng L, Rossi JM, Jain RK, Rao AV, Reagan PM (2020) KTE-X19 CAR T-cell therapy in relapsed or refractory mantle-cell lymphoma. *N Engl J Med* 382(14):1331–1342. <https://doi.org/10.1056/NEJMoa1914347>
- Xu Y, Yang Z, Horan LH, Zhang P, Liu L, Zimdahl B, Green S, Lu J, Morales JF, Barrett DM, Grupp SA, Chan VW, Liu H, Liu C (2018) A novel antibody-TCR (AbTCR) platform combines fab-based antigen recognition with gamma/delta-TCR signaling to facilitate T-cell cytotoxicity with low cytokine release. *Cell Discov* 4(1):62. <https://doi.org/10.1038/s41421-018-0066-6>
- Ying Z, Huang XF, Xiang X, Liu Y, Kang X, Song Y, Guo X, Liu H, Ding N, Zhang T, Duan P, Lin Y, Zheng W, Wang X, Lin N, Tu M, Xie Y, Zhang C, Liu W, Deng L, Gao S, Ping L, Wang X, Zhou N, Zhang J, Wang Y, Lin S, Mamuti M, Yu X, Fang L, Wang S, Song H, Wang G, Jones L, Zhu J, Chen SY (2019) A safe and potent anti-CD19 CAR T cell therapy. *Nat Med* 25(6):947–953. <https://doi.org/10.1038/s41591-019-0421-7>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.